

## Chương 6.

Exercise 5: Take the following Python code that stores a string: 'str = 'X-DSPAM-Confidence:0.8475'' Use find and string slicing to extract the portion of the string after the colon character and then use the float function to convert the extracted string into a floating point number.

```
C6 BT1 > ...
1  str = "X-DSPAM-Confidence:0.8475"
2  # tìm vị trí của dấu hai chấm
3  colon_pos = str.find(":")
4  # cắt chuỗi lấy phần sau hai dấu chấm
5  number_part = str[colon_pos + 1:]
6  # chuyển chuỗi thành số thực
7  value = float(number_part)
8  # in kết quả
9  print (value)
```

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Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLine for compatibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'.

PS F:\vscode> & C:/Users/tienp/AppData/Local/Programs/Python/Python313/python.exe "f:/vscode/C6 BT1"

0.8475

PS F:\vscode> & C:/Users/tienp/AppData/Local/Programs/Python/Python313/python.exe "f:/vscode/C6 BT1"

0.8475

PS F:\vscode>

## Chương 7:

Exercise 1: Write a program to read through a file and print the contents of the file (line by line) all in upper case. Executing the program will look as follows:

```
python shout.py
```

```
Enter a file name: mbox-short.txt
```

```
FROM STEPHEN.MARQUARD@UCT.AC.ZA SAT JAN 5 09:14:16  
2008
```

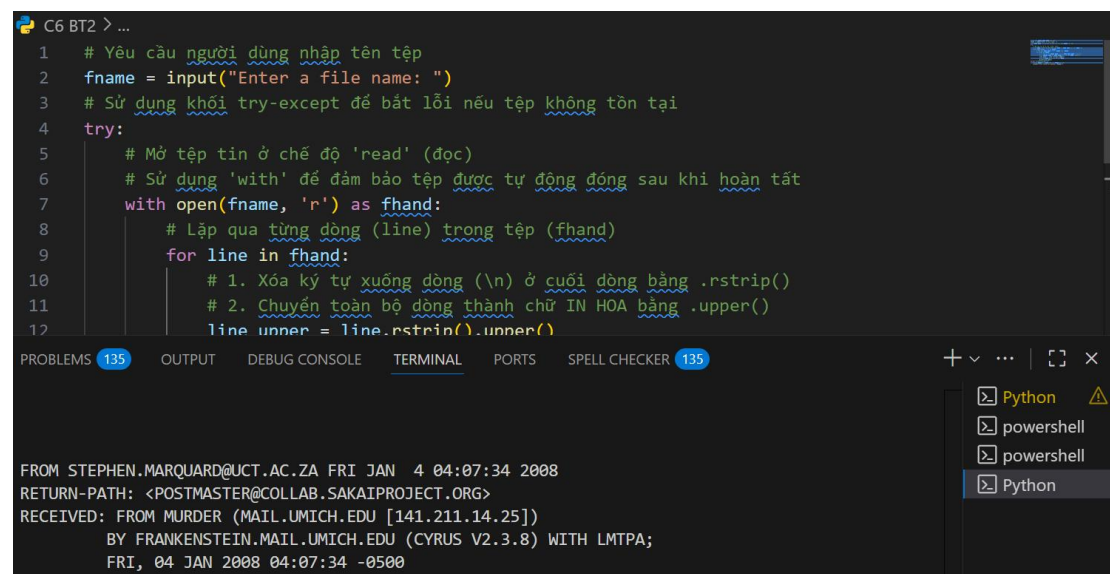
```
RETURN-PATH:
```

```
<POSTMASTER@COLLAB.SAKAIPROJECT.ORG>
```

```
RECEIVED: FROM MURDER (MAIL.UMICH.EDU [141.211.14.90])
```

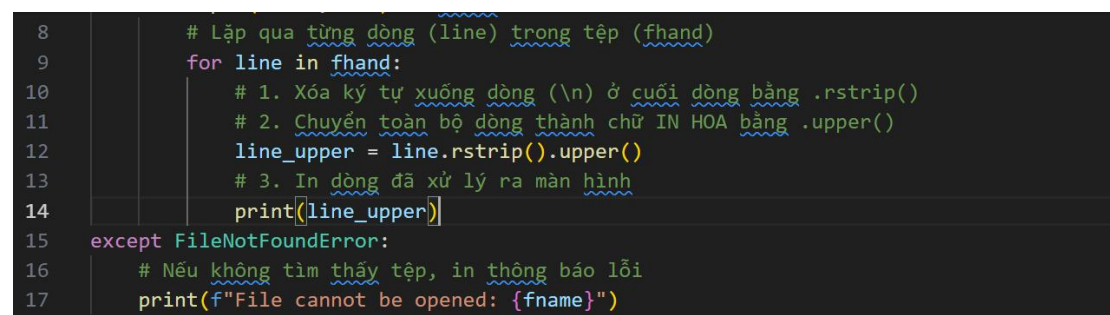
```
BY FRANKENSTEIN.MAIL.UMICH.EDU (CYRUS V2.3.8) WITH  
LMTPA;
```

```
SAT, 05 JAN 2008 09:14:16 -0500
```



The screenshot shows a code editor with a Python script and its output. The script is designed to read a file and print its contents in uppercase. The output shows the contents of the file 'mbox-short.txt' in uppercase.

```
1 # Yêu cầu người dùng nhập tên tệp
2 fname = input("Enter a file name: ")
3 # Sử dụng khối try-except để bắt lỗi nếu tệp không tồn tại
4 try:
5     # Mở tệp tin ở chế độ 'read' (đọc)
6     # Sử dụng 'with' để đảm bảo tệp được tự động đóng sau khi hoàn tất
7     with open(fname, 'r') as fhand:
8         # Lập qua từng dòng (line) trong tệp (fhand)
9         for line in fhand:
10             # 1. Xóa ký tự xuống dòng (\n) ở cuối dòng bằng .rstrip()
11             # 2. Chuyển toàn bộ dòng thành chữ IN HOA bằng .upper()
12             line_upper = line.rstrip().upper()
13
14 FROM STEPHEN.MARQUARD@UCT.AC.ZA FRI JAN 4 04:07:34 2008
15 RETURN-PATH: <POSTMASTER@COLLAB.SAKAIPROJECT.ORG>
16 RECEIVED: FROM MURDER (MAIL.UMICH.EDU [141.211.14.25])
17 BY FRANKENSTEIN.MAIL.UMICH.EDU (CYRUS V2.3.8) WITH LMTPA;
18 FRI, 04 JAN 2008 04:07:34 -0500
```



The screenshot shows a code editor with a Python script. The script is designed to read a file and print its contents in uppercase. The script includes comments in Vietnamese and handles a FileNotFoundError exception.

```
8     # Lập qua từng dòng (line) trong tệp (fhand)
9     for line in fhand:
10         # 1. Xóa ký tự xuống dòng (\n) ở cuối dòng bằng .rstrip()
11         # 2. Chuyển toàn bộ dòng thành chữ IN HOA bằng .upper()
12         line_upper = line.rstrip().upper()
13         # 3. In dòng đã xử lý ra màn hình
14         print(line_upper)
15 except FileNotFoundError:
16     # Nếu không tìm thấy tệp, in thông báo lỗi
17     print(f"File cannot be opened: {fname}")
```

Exercise 2: Write a program to prompt for a file name, and then read through the file and look for lines of the form: X-DSPAM-Confidence:0.8475 When you encounter a line that starts with “X-DSPAM-Confidence:” pull apart the line to extract the floating-point number on the line. Count these lines and then compute the total of the spam confidence values from these lines. When you reach the end of the file, print out the average spam confidence. Enter the file name: mbox.txt Average spam confidence: 0.894128046745 Enter the file name: mbox-short.txt Average spam confidence: 0.750718518519 Test your file on the mbox.txt and mbox-short.txt files.

```

C7 BT3 > ...
1  # Nhập tên file từ bàn phím
2  fname = input("Enter the file name: ")
3  # Mở file để đọc
4  fh = open(fname)
5  count = 0      # đếm số dòng có "X-DSPAM-Confidence"
6  total = 0.0    # tổng giá trị confidence
7  for line in fh:
8      line = line.strip() # bỏ khoảng trắng đầu/cuối
9      if line.startswith("X-DSPAM-Confidence:"):
10         # Tách lấy phần số sau dấu hai chấm
11         parts = line.split(":")
12         number = float(parts[1].strip()) # chuyển sang dạng float
13
14
15     # Tách lấy phần số sau dấu hai chấm
16     parts = line.split(":")
17     number = float(parts[1].strip()) # chuyển sang dạng float
18     total += number
19     count += 1
20
21 # Tính trung bình
22 if count > 0:
23     average = total / count
24     print("Average spam confidence:", average)
25 else:
26     print("Không tìm thấy dòng nào có 'X-DSPAM-Confidence'")
27

```

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```

ode/C7 BT3"
Traceback (most recent call last):
  File "f:\vscode\C7 BT3", line 5, in <module>
    fh = open(fname)
OSError: [Errno 22] Invalid argument: '& C:/Users/tienp/AppData/Local/Programs/Python/Python313/python.exe "f:/vscode/C7 BT3"'
PS F:\vscode> & C:/Users/tienp/AppData/Local/Programs/Python/Python313/python.exe "f:/vscode/C7 BT3"
Enter the file name: mbox-short.txt
Average spam confidence: 0.7507185185187
PS F:\vscode>

```

Exercise 3: Sometimes when programmers get bored or want to have a bit of fun, they add a harmless Easter Egg to their program. Modify the program that prompts the user for the file name so that it prints a funny message when the user types in the exact file name “na na boo boo”. The program should behave normally for all other files which exist and don’t exist. Here is a sample execution of the program: 90 .

```
Fpython egg.py
```

```
Enter the file name: mbox.txt There were 1797 subject lines in
mbox.txt
```

```
python egg.py
```

```
Enter the file name: missing.tyxt
```

```
File cannot be opened: missing.tyxt
```

```
python egg.py
```

```
Enter the file name: na na boo boo NA NA BOO BOO TO YOU - You have
been punk'd!
```

We are not encouraging you to put Easter Eggs in your programs; this is just an exercise.

```
1  # Nhập tên file từ bàn phím
2  fname = input("Enter the file name: ")
3  # Kiểm tra nếu người dùng nhập 'na na boo boo'
4  if fname.lower() == "na na boo boo":
5      print("NA NA BOO BOO TO YOU - You have been punk'd!")
6      quit() # kết thúc chương trình ở đây
7  # Cố gắng mở file, nếu không có thì báo lỗi
8  try:
9      fhand = open(fname)
10 except:
11     print("File cannot be opened:", fname)
12     quit()
```

```

10 except:
11     print("File cannot be opened:", fname)
12     quit()
13 # Đếm số dòng bắt đầu bằng 'Subject:'
14 count = 0
15 for line in fhand:
16     if line.startswith("Subject:"):
17         count += 1
18 # In kết quả ra màn hình
19 print("There were", count, "subject lines in", fname)
20

```

PROBLEMS 172

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER 172

BT"

Enter the file name: mbox-short.txt

There were 27 subject lines in mbox-short.txt

PS F:\vscode> & C:/Users/tienp/AppData/Local/Programs/Python/Python313/python.exe "f:/vscode/C7

BT"

Enter the file name: na na boo boo

NA NA BOO BOO TO YOU - You have been punk'd!

PS F:\vscode> & C:/Users/tienp/AppData/Local/Programs/Python/Python313/python.exe "f:/vscode/C7

BT"

Enter the file name: missing.tyxt

File cannot be opened: missing.tyxt